MUSCATINE, IA, USA

WMO: **725487**

Lat: **41.367N** Lon: **91.150W** Elev: **167** StdP: **99.34** Time Zone: **-6.00 (NAC)** Period: **95-19** WBAN: **04950**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-19.1	-16.3	-24.1	0.4	-18.1	-22.1	0.5	-15.5	13.0	-3.4	11.7	-4.5	3.3	300	0.511	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	11.6	34.9	25.2	32.9	24.3	32.0	23.8	27.0	32.2	25.9	31.2	25.0	29.8	4.4	200	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	25.8	21.6	30.3	24.6	20.0	29.2	23.0	18.1	27.8	85.9	32.4	81.2	31.1	76.6	30.2	34.6	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	11.1	9.4	8.4	DB	-25.0	37.0	5.8	2.3	-29.2	38.6	-32.6	39.9	-35.8	41.2	-40.1	42.8	(4)
(5)				WB	-24.7	28.9	5.4	2.0	-28.5	30.4	-31.7	31.6	-34.7	32.8	-38.6	34.2	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	11.0	-4.3	-2.2	4.2	11.0	17.1	22.5	24.7	23.2	19.4	12.2	4.6	-1.5	(6)
(7)			DBStd	11.40	6.68	6.59	6.45	5.42	4.70	3.61	3.42	3.04	4.22	5.14	5.75	6.08	(7)
(8)			HDD10.0	1617	442	344	202	51	3	0	0	0	0	36	178	360	(8)
(9)			HDD18.3	3296	700	576	440	225	81	7	1	2	37	199	412	616	(9)
(10)			CDD10.0	1978	0	2	22	82	224	376	457	409	283	104	17	2	(10)
(11)			CDD18.3	616	0	0	1	6	44	133	199	153	70	9	1	0	(11)
(12)			CDH23.3	6307	0	0	16	98	436	1315	2166	1419	734	113	4	6	(12)
(13)			CDH26.7	2508	0	0	2	20	130	531	980	546	269	26	0	4	(13)
(14)	Wind		WSAvg	3.4	3.9	3.9	4.1	4.3	3.7	3.0	2.5	2.2	2.5	3.3	3.8	3.6	(14)
(15)	Precipitation		PrecAvg	975	36	42	69	93	126	141	107	111	80	72	52	46	(15)
(16)			PrecMax	1406	70	108	150	165	307	303	291	298	211	182	142	96	(16)
(17)			PrecMin	534	8	10	14	31	24	41	15	24	15	13	4	13	(17)
(18)			PrecStd	225	16	27	45	36	64	76	68	66	53	48	36	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		13.8	17.6	26.1	29.2	32.6	35.9	37.9	35.9	34.1	29.9	23.6	17.7	(19)
(20)			MCWB		8.2	11.8	17.8	18.4	22.4	25.2	25.6	25.5	23.5	20.6	16.4	14.8	(20)
(21)		2%	DB		9.8	12.8	21.2	27.0	29.9	33.6	35.2	32.9	32.3	26.4	19.2	12.7	(21)
(22)			MCWB		7.0	8.7	14.5	17.5	20.8	23.9	26.2	25.1	22.7	17.5	13.6	9.9	(22)
(23)		5%	DB		6.4	9.1	17.5	23.3	27.7	32.0	33.0	31.4	30.0	23.7	17.0	9.1	(23)
(24)			MCWB		3.8	6.2	12.4	15.2	19.4	22.9	25.0	24.2	21.7	16.5	12.2	7.0	(24)
(25)		10%	DB		3.8	6.3	13.9	21.0	25.9	29.9	32.1	29.9	27.6	21.2	13.7	7.0	(25)
(26)			MCWB		2.1	3.7	9.4	13.9	18.5	21.7	24.5	23.3	20.1	15.4	9.6	4.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		9.9	13.1	18.6	20.7	24.4	27.0	28.9	28.4	25.7	22.4	17.5	15.7	(27)
(28)			MCDB		11.3	15.6	24.1	26.8	30.0	33.2	34.0	32.7	31.3	27.5	22.6	17.8	(28)
(29)		2%	WB		6.5	9.4	15.7	18.7	22.4	25.5	27.4	26.7	24.2	20.3	15.0	10.7	(29)
(30)			MCDB		9.2	13.0	19.9	24.2	27.8	31.4	32.5	31.3	29.9	23.4	18.2	12.0	(30)
(31)		5%	WB		4.2	6.3	12.8	16.8	21.2	24.2	26.3	25.6	23.1	18.1	12.7	7.4	(31)
(32)			MCDB		6.6	9.4	16.3	21.9	25.9	29.6	32.0	29.7	27.5	21.5	16.1	9.0	(32)
(33)		10%	WB		2.4	3.9	9.9	14.9	19.7	23.0	25.3	24.4	21.8	16.3	10.1	4.5	(33)
(34)			MCDB		4.0	6.2	14.0	20.1	24.3	28.1	30.6	28.2	26.6	20.6	13.3	6.6	(34)
(35)	Mean Daily Temperature Range		MDBR		8.7	9.1	10.5	12.1	11.5	11.4	11.6	11.5	13.2	12.0	10.2	8.3	(35)
(36)		5% DB	MCDBR		11.8	13.5	15.1	16.5	13.8	13.6	13.1	13.4	15.4	15.9	14.6	12.2	(36)
(37)			MCWBR		9.2	9.6	9.4	9.3	6.9	6.1	6.0	6.6	7.2	8.5	9.8	9.8	(37)
(38)		5% WB	MCDBR		11.5	13.1	13.4	13.7	12.0	11.9	11.7	11.3	12.4	11.7	13.6	11.3	(38)
(39)			MCWBR		9.0	9.7	9.2	9.0	7.1	6.4	6.2	6.4	6.5	7.9	10.1	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.292	0.307	0.342	0.388	0.430	0.428	0.445	0.428	0.396	0.348	0.305	0.293	(40)
(41)		taud			2.422	2.390	2.378	2.259	2.205	2.266	2.199	2.258	2.309	2.443	2.496	2.437	(41)
(42)		Ebn at Noon			864	904	906	881	849	850	830	834	837	845	845	831	(42)
(43)		Edh at Noon			79	95	108	131	142	134	142	130	115	90	73	72	(43)
(44)	All-Sky Solar Radiation	RadAvg			1.80	2.78	3.75	4.71	5.52	6.17	6.15	5.42	4.61	3.05	2.04	1.49	(44)
(45)		RadStd			0.21	0.25	0.31	0.39	0.40	0.58	0.42	0.34	0.37	0.34	0.17	0.17	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page